

PREDICTING WINE QUALITY:

A Comparison of Linear Regression Techniques on a
Multicollinear and Outlier-Affected Dataset

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Semester 6,

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2 May, 2025

Abstract

➤ **Goal:** Predict wine quality based on 10 physicochemical features (e.g., pH, sulphur dioxide, anthocyanins)

➤ **Data:** Small, real-world dataset ($n = 32$), with multicollinearity and numerous outliers

➤ **Methods Used:**

- Ordinary Least Squares (OLS)
- Stepwise Regression (both-way)
- Robust Regression (Huber's M-estimator & MM-estimator)

➤ **Challenges Addressed:**

- Multicollinearity (checked via VIF)
- Outliers
- Heteroscedasticity

➤ **Key Finding:**

MM-estimator offered the best trade-off between fit and robustness

➤ **Conclusion:**

Robust methods are essential for noisy, small-sample predictive modeling

Introduction

Why is Wine Quality Important?

- Quality determines consumer preference, pricing, and production choices
- Traditionally assessed by sommeliers, but subjective & inconsistent

What Factors Affect Wine Quality?

- Physicochemical properties (pH, sulphur dioxide, anthocyanin, colour density)

Opportunity

- Using physicochemical properties we can predict quality objectively & scientifically

Few Past Researches

- Dahal et al. (2021): Ridge, SVM, Gradient Boosting demonstrated improved performance especially in handling high-dimensional data
- Arshad (2024): Random Forest, XGBoost → 95%+ accuracy on larger, well-curated datasets

The Missing Piece

- Many ignore multicollinearity & outliers which are inevitable
- Real-world data $\neq$ perfect data

Introduction (Cont.)

Our Approach:

- Dataset Overview:

A real-world, modest-sized dataset containing 32 wine samples, each described by 10 physicochemical features—including pH, sulphur dioxide levels, anthocyanin concentration, and colour density

- Initial Challenges:

-  Multicollinearity – Several predictors are highly correlated, which can distort model coefficients and reduce interpretability
-  Outliers – Extreme values may skew predictions, making many models unreliable without proper handling

- Our Solution: We adopt a robust, adaptive modelling framework that addresses these issues head-on. By incorporating techniques like feature selection and outlier-resistant algorithms, we aim to build models that perform well not just on paper—but in messy, real-world scenarios where data isn't perfect

Objectives of the Study

Goal: Develop a systematic & objective approach to wine quality assessment

How?

- Begin with EDA to understand the nature of the dataset
- Use regression analysis to model wine quality based on physicochemical attributes
- Analyze wine samples using the following features:
 - Wine varietal (Cabernet Sauvignon = 0, Shiraz = 1)
 - pH levels, total SO₂ (ppm)
 - Colorimetric properties (color density, wine color, polymeric pigment color)
 - Anthocyanin properties (total anthocyanins, ionization degree, ionized anthocyanins)

Outcome:

- Identify the most affective model which can explain variability in wine quality ratings
- Provide a data-driven approach for better quality control & consistency

Data Source & Description

Repository: Originally sourced from the UCI Machine Learning Repository (now archived and no longer publicly available)

Dataset was provided to me by my professor for academic use; previously downloaded and preserved earlier

Parameters:

y: quality rating (20 maximum)
x_1: wine varietal (0 - Cabernet Sauvignon, 1 - Shiraz)
x_2: pH
x_3: Total SO₂(ppm)
x_4: color density
x_5: wine color
x_6: polymeric pigment color
x_7: anthocyanin color
x_8: total anthocyanins (g/L)
x_9: degree of ionization of anthocyanins (percent)
x_10: ionized anthocyanins (percent)

Dataset

Train Data:

	y	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_10
1	19.2	0	3.85	66	9.35	5.65	2.40	3.25	0.33	19	0.065
2	18.3	0	3.73	79	11.15	6.95	3.15	3.80	0.36	21	0.076
3	17.1	0	3.88	73	9.40	5.75	2.10	3.65	0.40	18	0.073
5	16.8	0	3.98	75	8.55	5.05	2.05	3.00	0.49	12	0.060
8	15.2	0	3.66	86	6.40	4.00	1.50	2.50	0.27	19	0.050
9	15.2	0	3.91	78	5.80	3.30	1.40	1.90	0.40	9	0.038
10	14.0	0	3.47	178	3.60	2.25	0.75	1.50	0.37	8	0.030
11	14.0	0	3.91	81	3.90	2.15	1.00	1.15	0.32	7	0.023
12	13.8	0	3.75	108	5.80	3.20	1.60	1.60	0.38	8	0.032
13	13.6	0	3.90	92	5.40	2.85	1.55	1.30	0.44	6	0.026
14	12.8	0	3.92	96	5.00	2.70	1.40	1.30	0.35	7	0.026
15	18.5	1	3.87	89	9.15	5.60	1.95	3.65	0.46	16	0.073
16	17.3	1	3.97	59	10.25	6.10	2.40	3.70	0.40	19	0.074
17	16.3	1	3.76	22	8.20	5.00	1.85	3.15	0.25	25	0.063
19	16.0	1	3.98	58	10.15	6.00	2.60	3.40	0.38	18	0.068
20	16.0	1	3.88	85	6.85	4.10	1.50	2.60	0.33	16	0.052
22	15.5	1	3.98	94	5.45	3.05	1.50	1.55	0.41	8	0.031
23	15.3	1	3.69	122	8.00	5.05	1.90	3.15	0.27	23	0.063
24	15.3	1	3.77	144	5.60	3.35	1.10	2.25	0.36	12	0.045
25	14.8	1	3.74	10	7.90	4.75	1.95	2.80	0.25	23	0.056
26	14.3	1	3.76	100	5.55	3.25	1.15	2.10	0.34	12	0.042
27	14.3	1	3.91	73	4.65	2.70	0.95	1.75	0.36	10	0.035
28	14.2	1	3.60	301	4.25	2.40	1.25	1.15	0.42	6	0.023
30	13.8	1	3.90	67	7.40	4.40	1.60	2.80	0.45	13	0.056
31	12.5	1	3.80	89	5.35	3.15	1.20	1.95	0.32	12	0.039
32	11.5	1	3.65	192	6.35	3.90	1.25	2.65	0.63	8	0.053

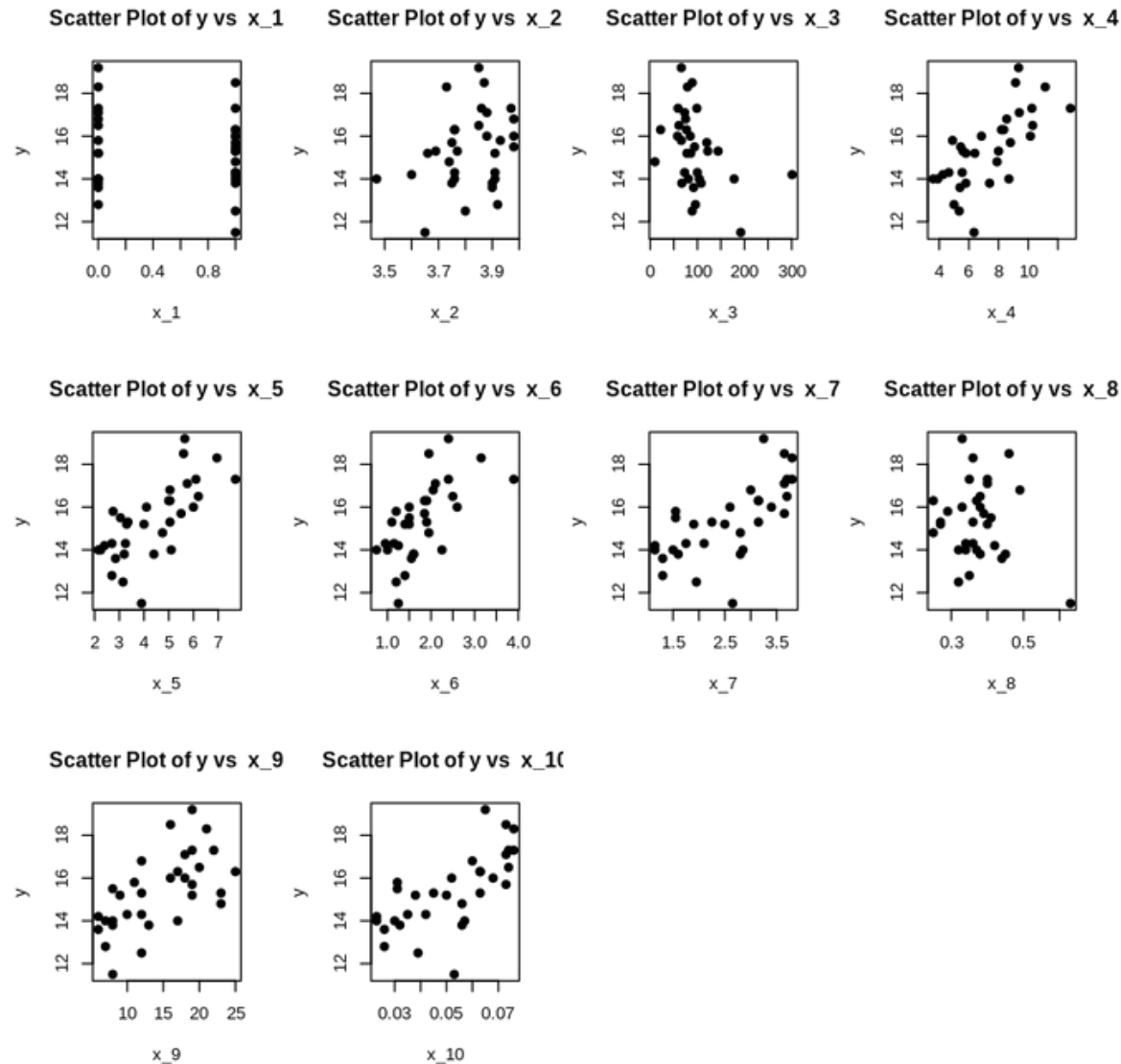
Dataset (Cont.)

Test Data:

	y	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_10
4	17.3	0	3.86	99	12.85	7.70	3.90	3.80	0.35	22	0.076
6	16.5	0	3.85	61	10.30	6.20	2.50	3.70	0.38	20	0.074
7	15.8	0	3.93	66	4.90	2.75	1.20	1.55	0.29	11	0.031
18	16.3	1	3.76	77	8.35	5.05	1.90	3.15	0.37	17	0.063
21	15.7	1	3.75	120	8.80	5.50	1.85	3.65	0.39	19	0.073
29	14.0	1	3.76	104	8.70	5.10	2.25	2.85	0.34	17	0.057

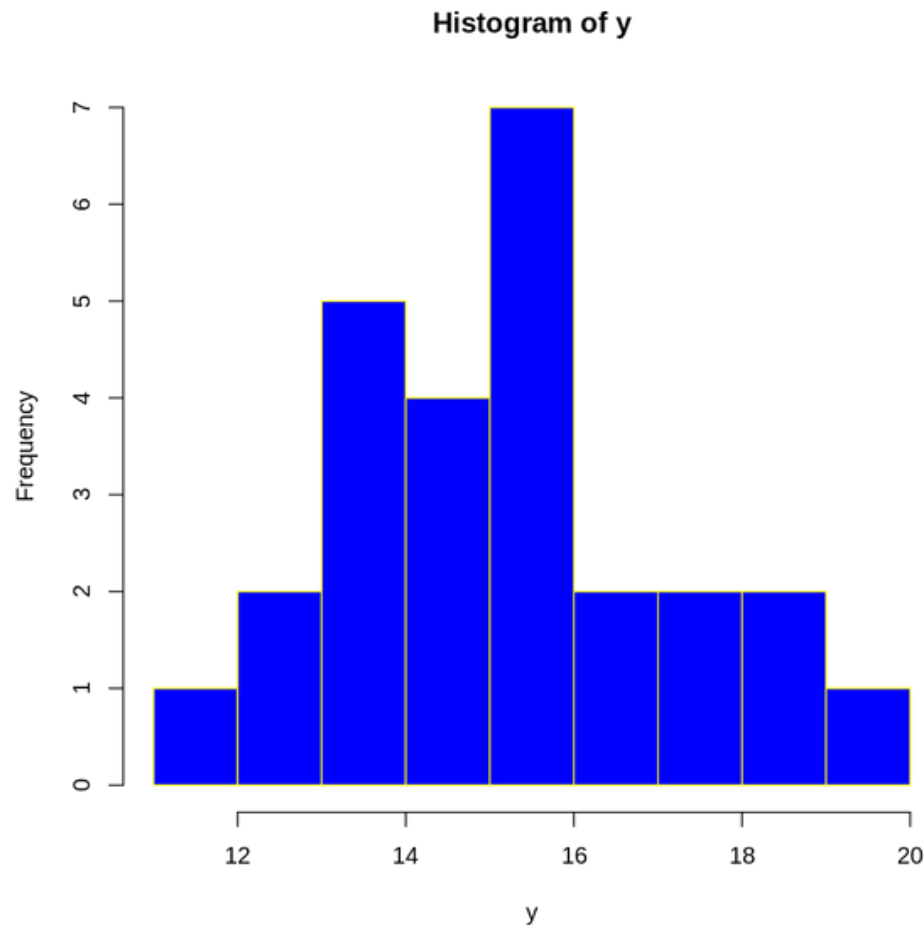
Methodology – Exploratory Data Analysis

Scatter Plots: To visualise relationships between individual features and wine quality



Methodology – Exploratory Data Analysis (Cont.)

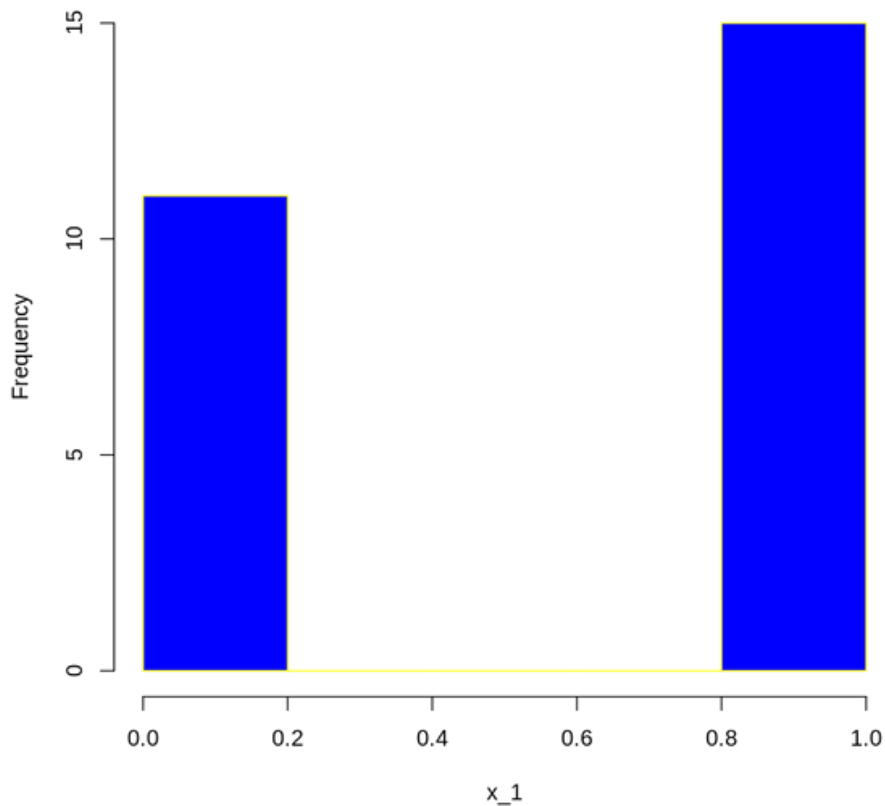
Histograms: To observe the distribution and skewness of each variable



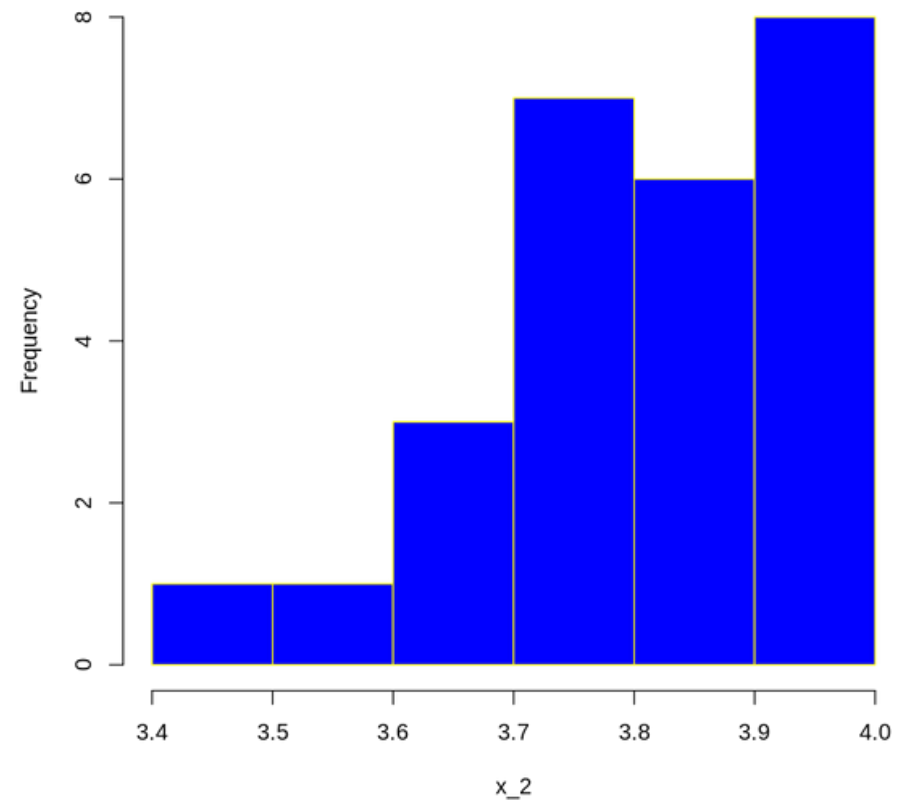
Methodology – Exploratory Data Analysis (Cont.)

Histograms:

Histogram of x_1



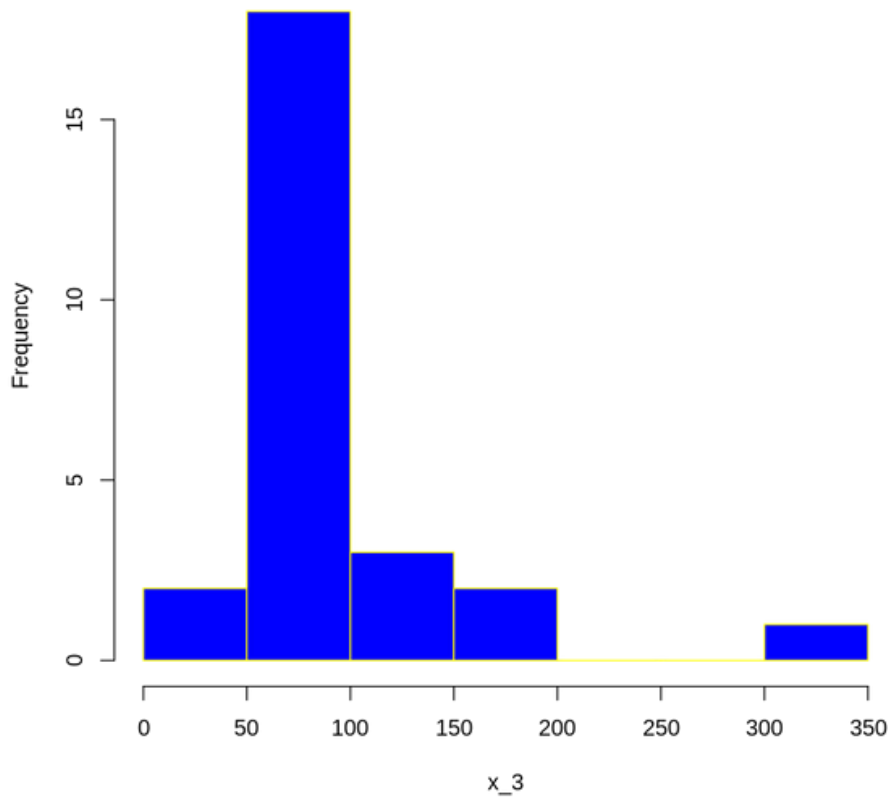
Histogram of x_2



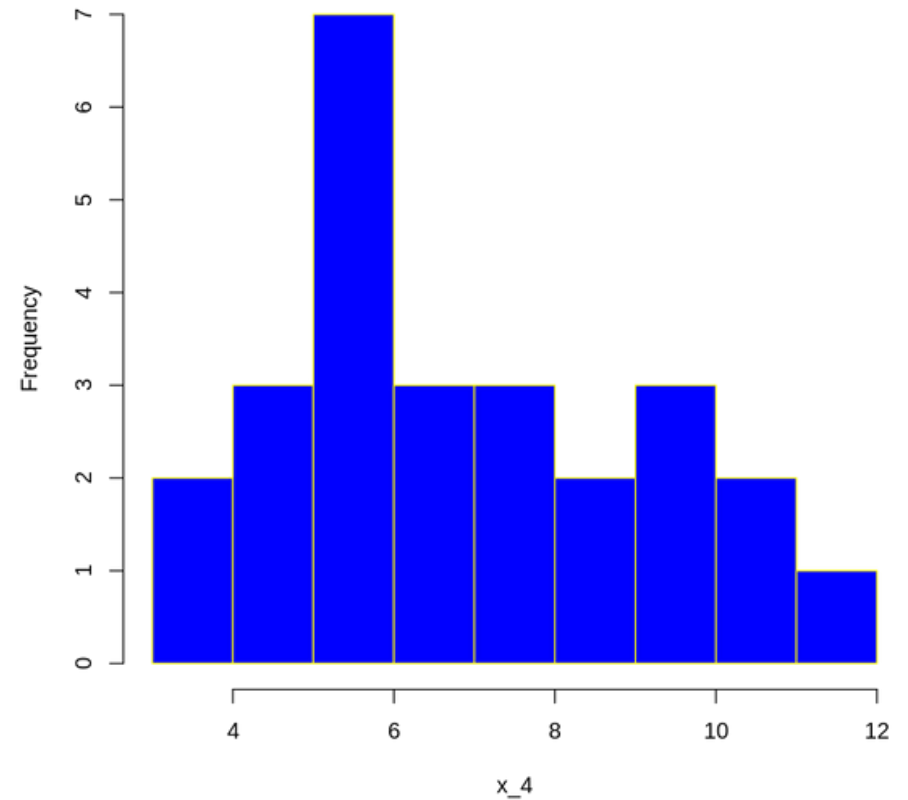
Methodology – Exploratory Data Analysis (Cont.)

Histograms

Histogram of x_3



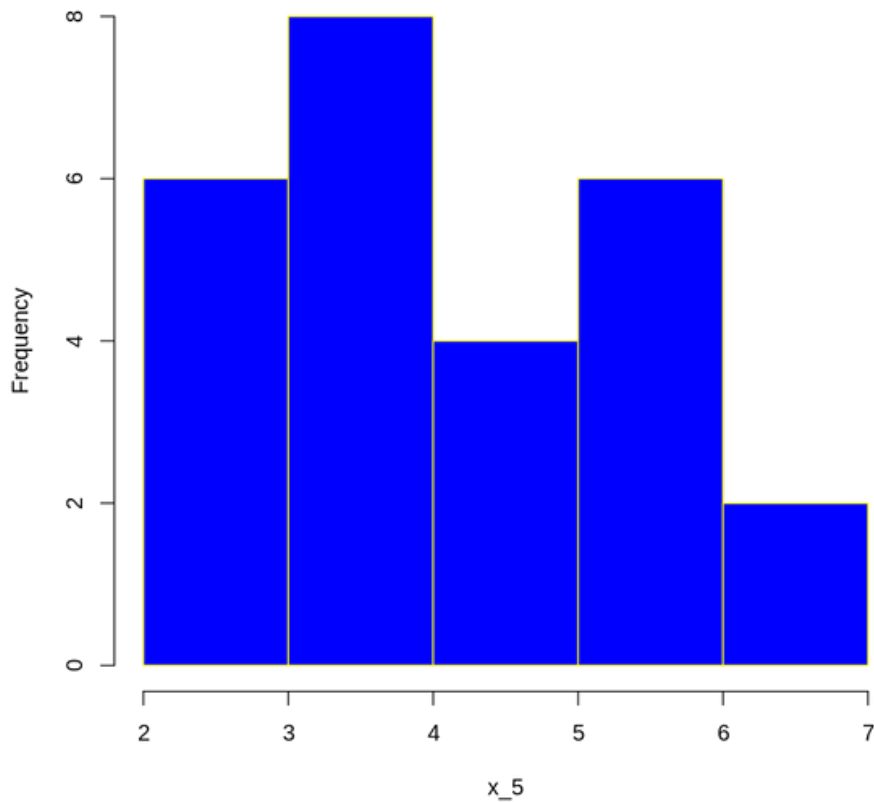
Histogram of x_4



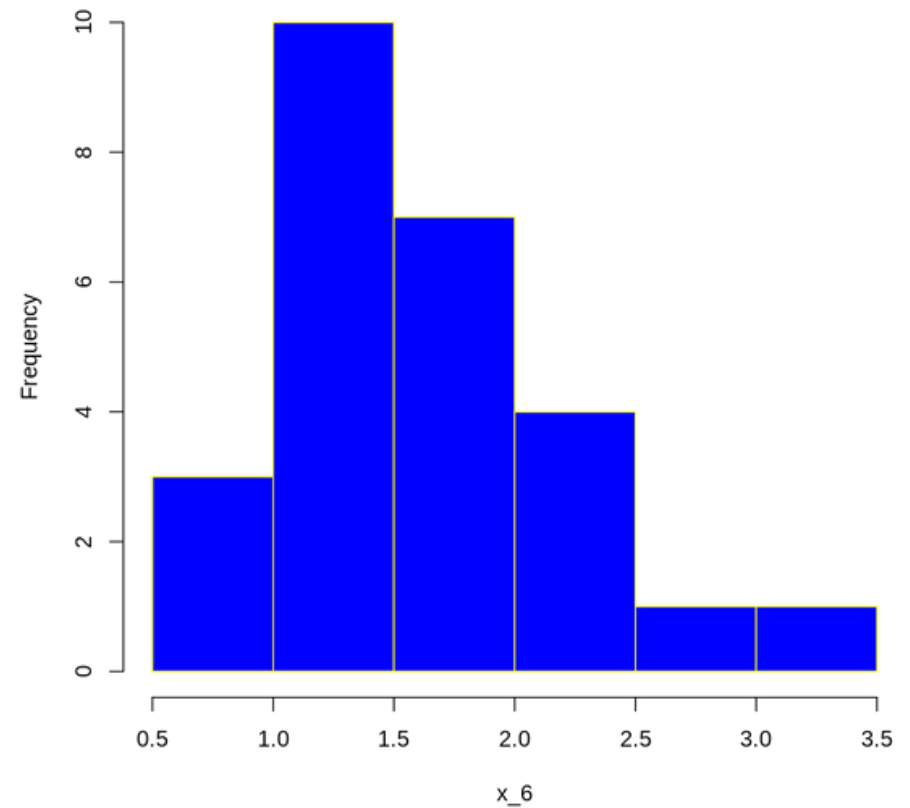
Methodology – Exploratory Data Analysis (Cont.)

Histograms

Histogram of x_5



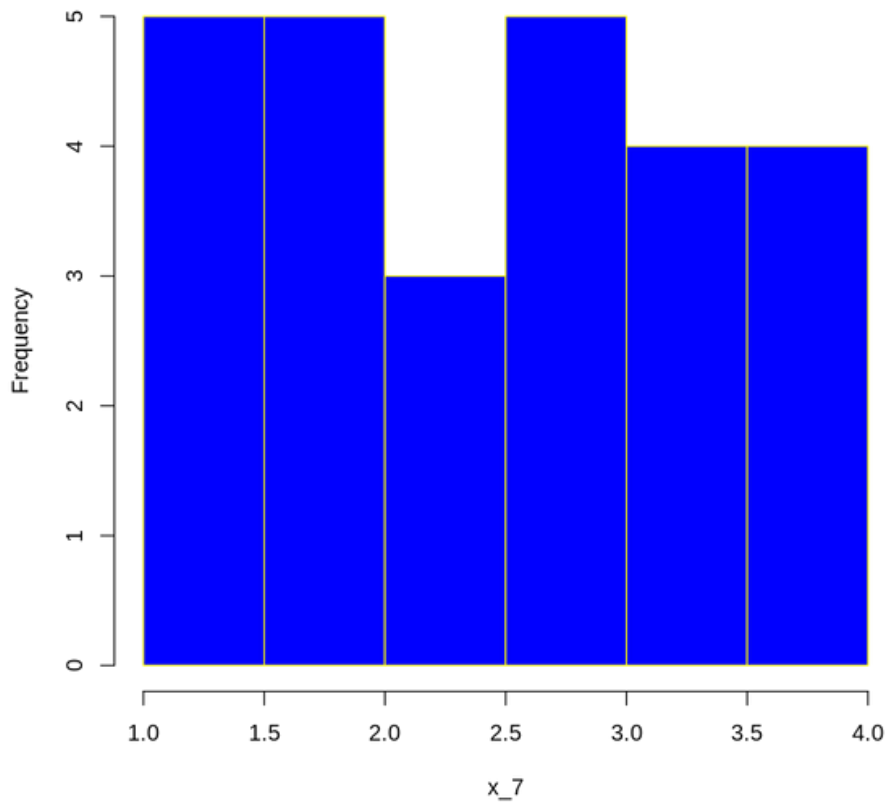
Histogram of x_6



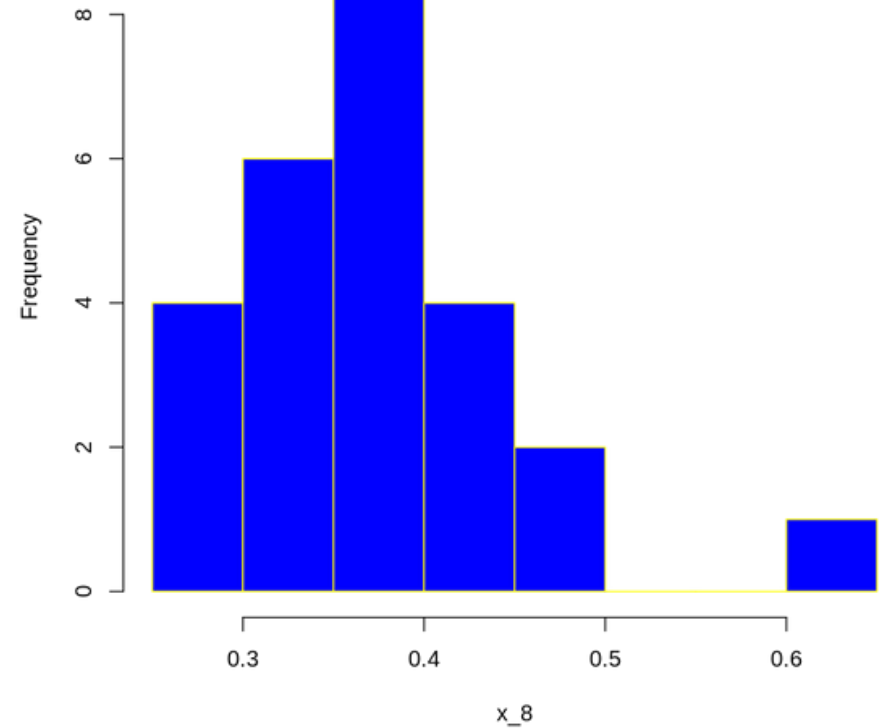
Methodology – Exploratory Data Analysis (Cont.)

Histograms

Histogram of x_7



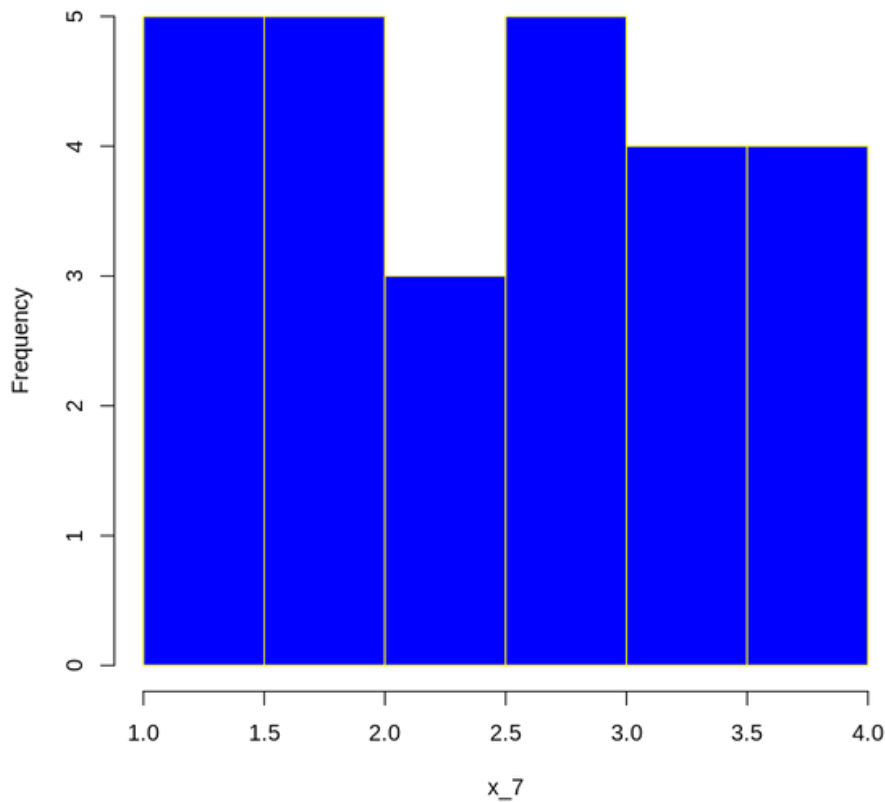
Histogram of x_8



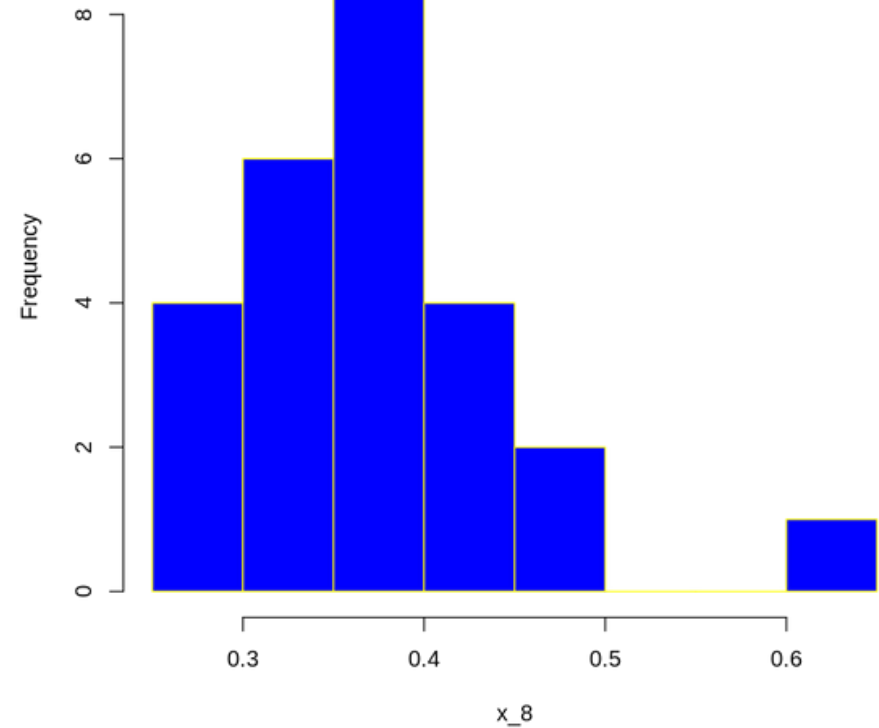
Methodology – Exploratory Data Analysis (Cont.)

Histograms

Histogram of x_7



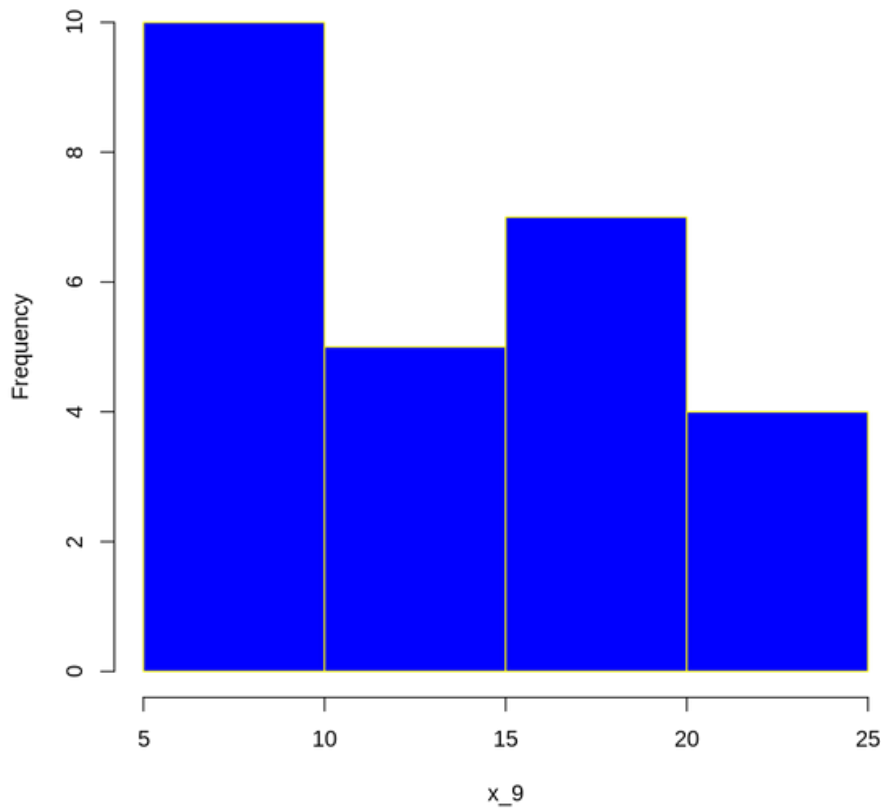
Histogram of x_8



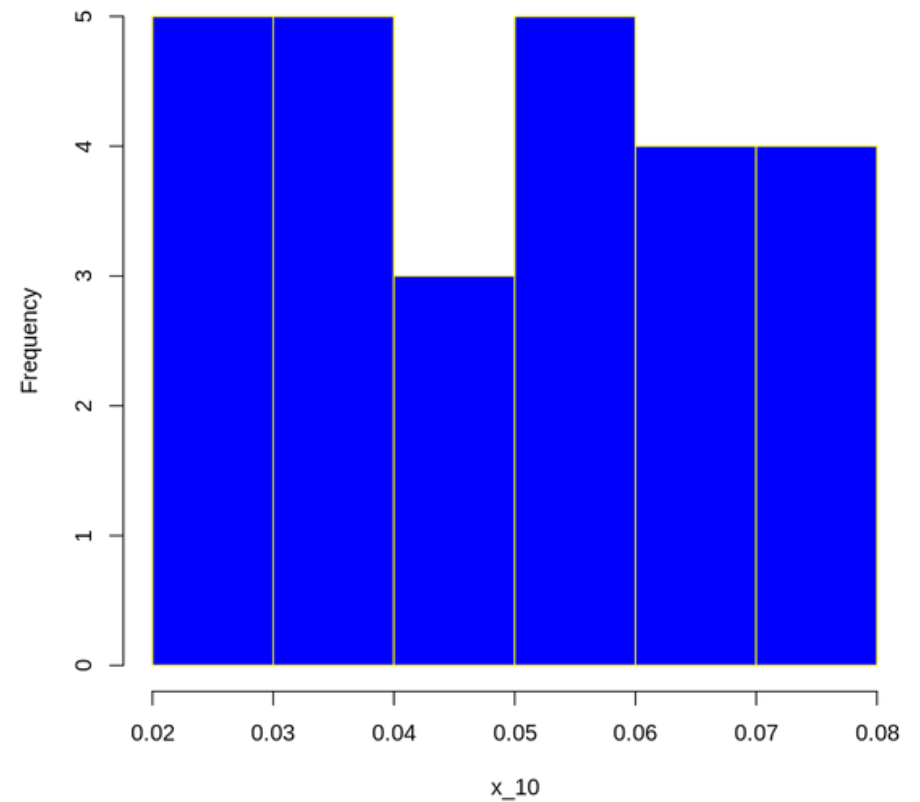
Methodology – Exploratory Data Analysis (Cont.)

Histograms

Histogram of x_9

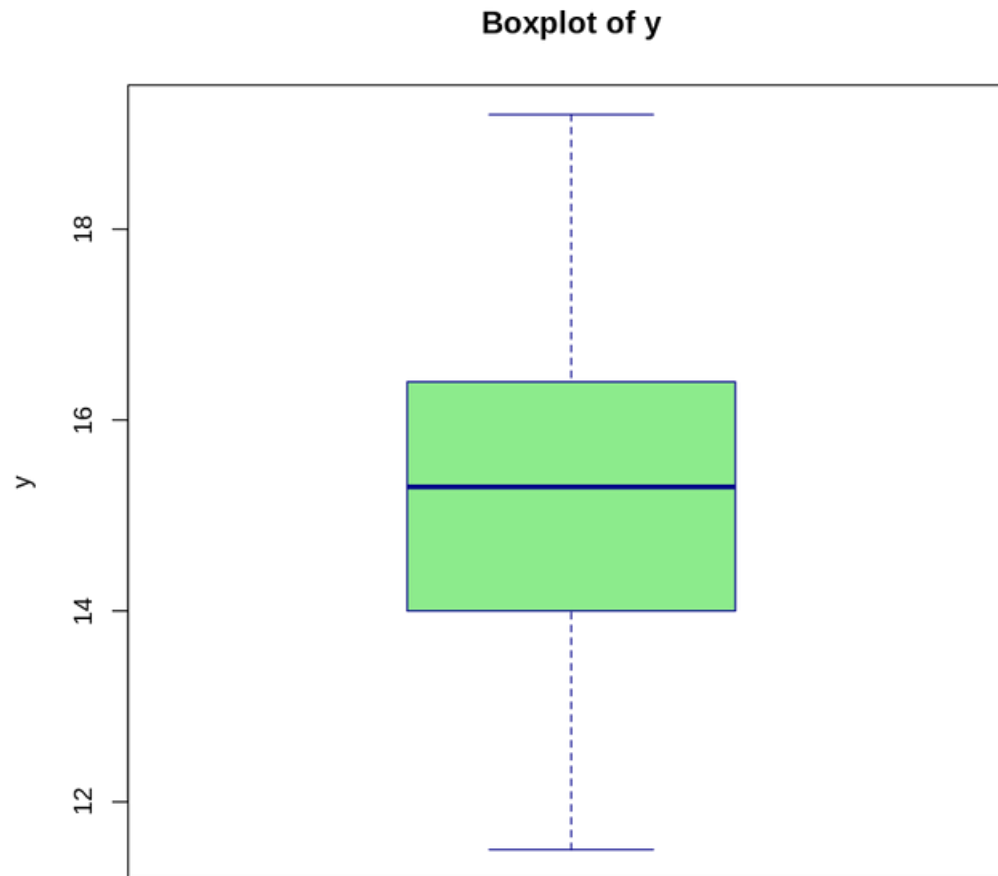


Histogram of x_10



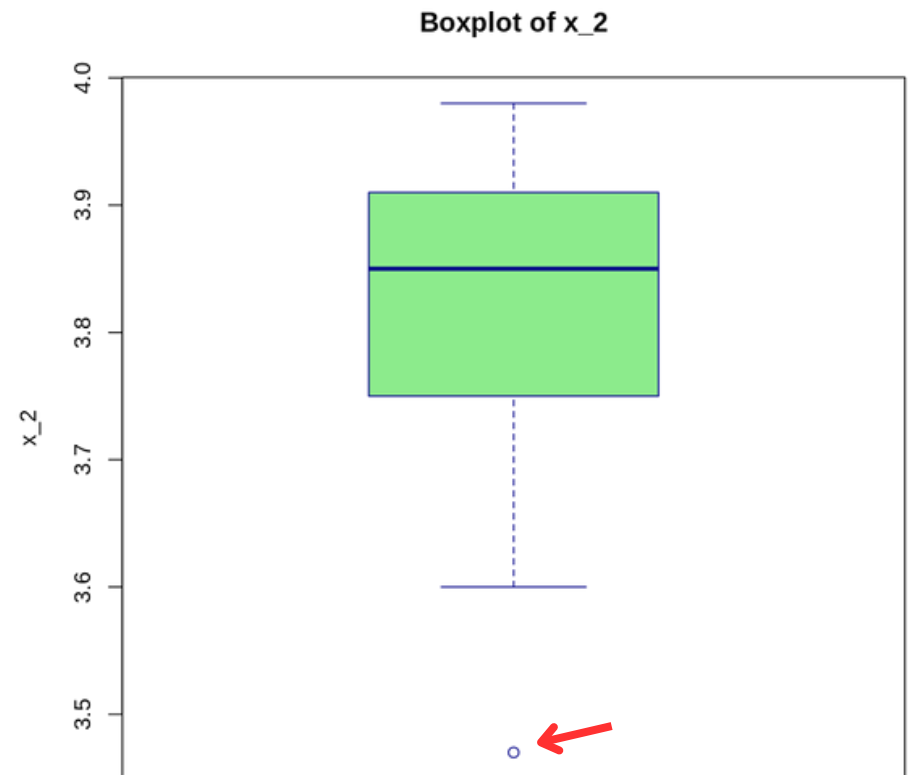
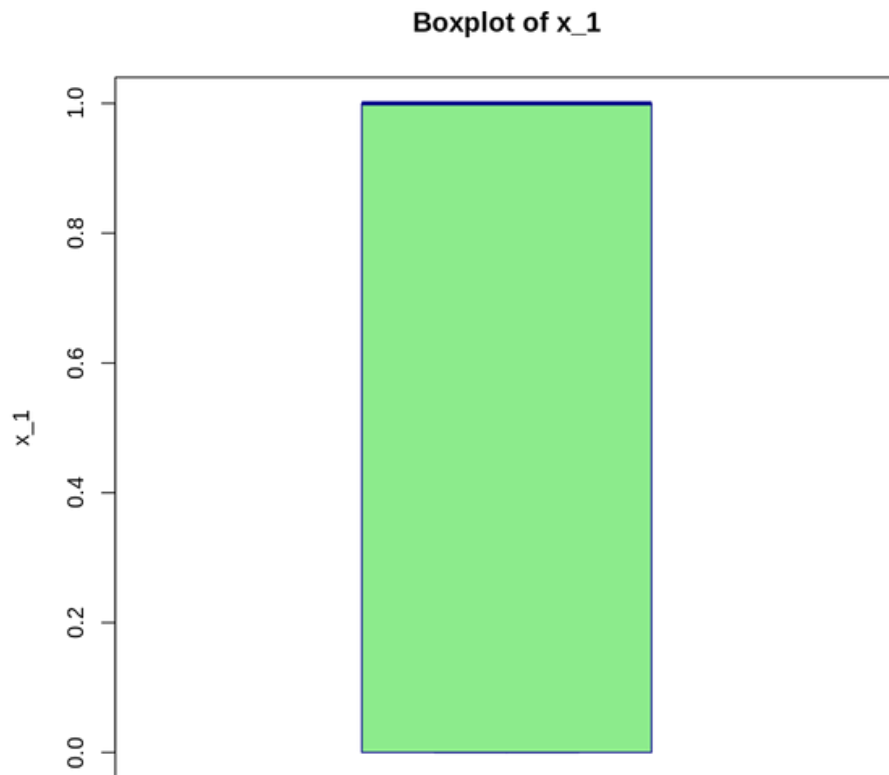
Methodology – Exploratory Data Analysis (Cont.)

Box Plot: To detect outliers and compare feature spread across different samples



Methodology – Exploratory Data Analysis (Cont.)

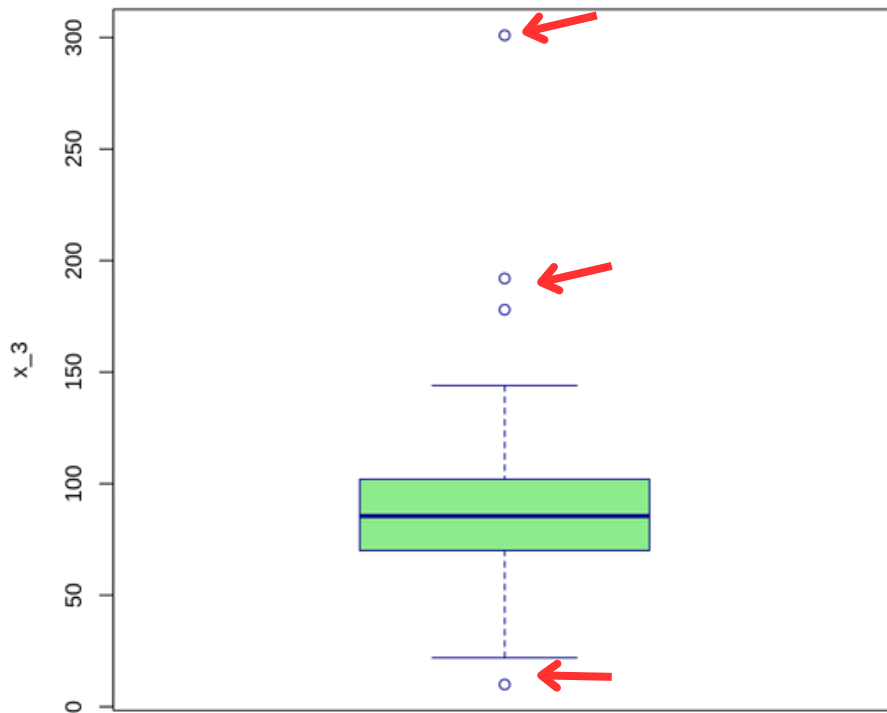
Box Plots



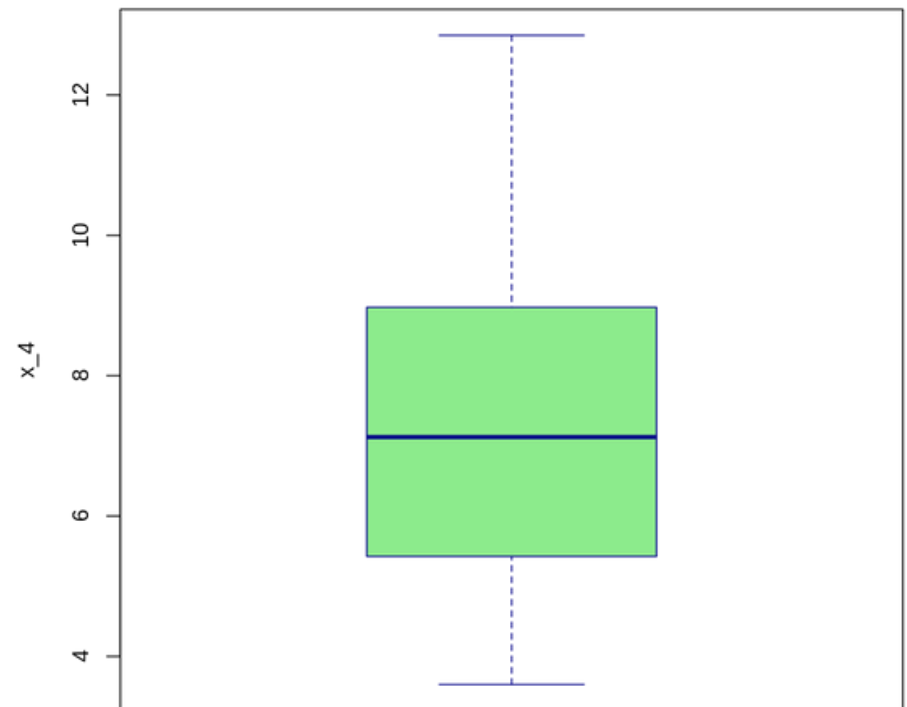
Methodology – Exploratory Data Analysis (Cont.)

Box Plots

Boxplot of x_3



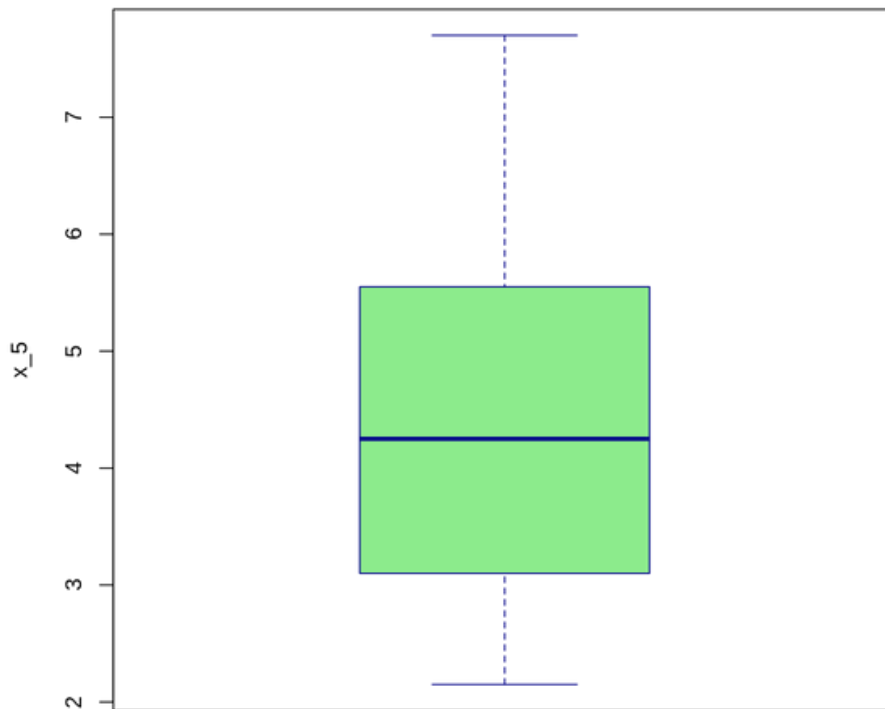
Boxplot of x_4



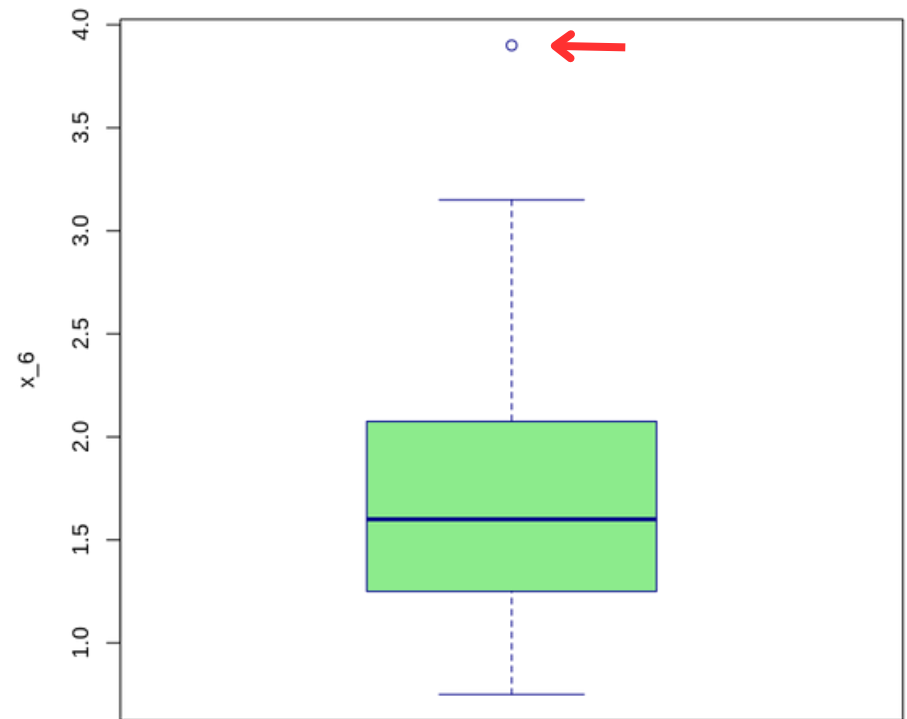
Methodology – Exploratory Data Analysis (Cont.)

Box Plots

Boxplot of x_5



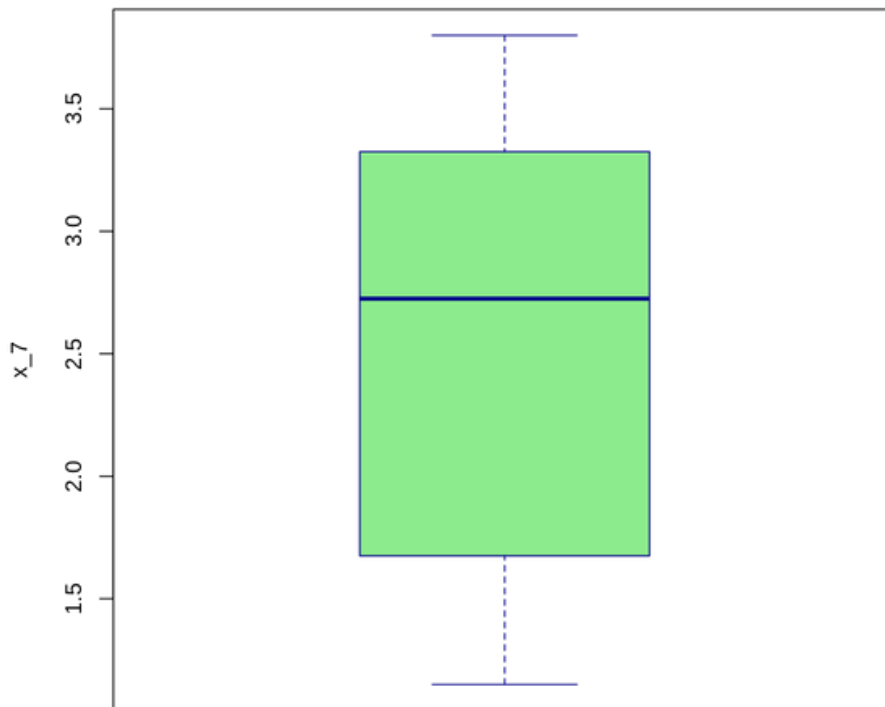
Boxplot of x_6



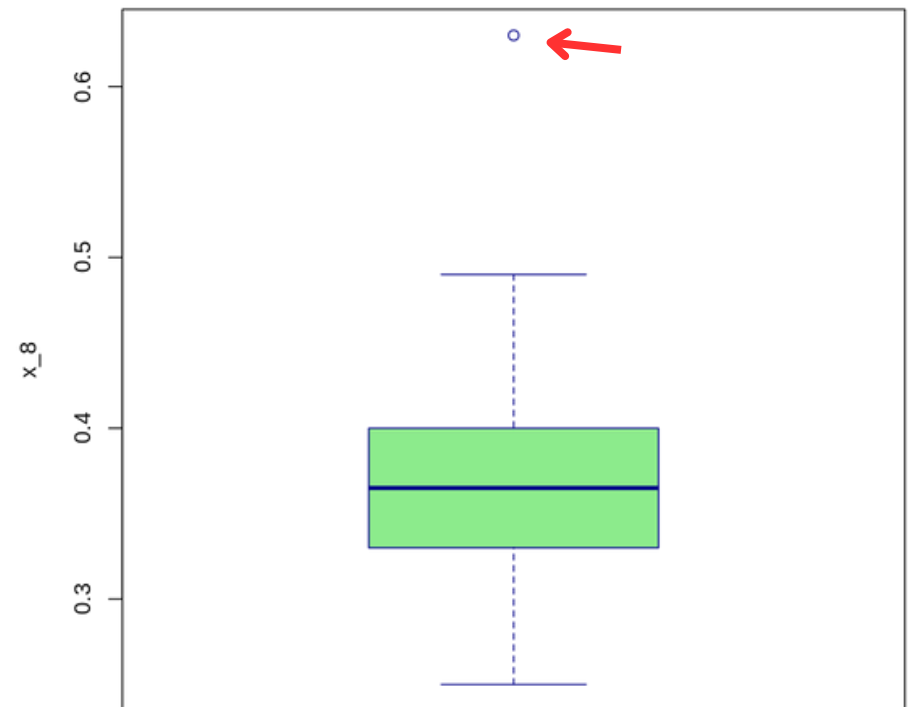
Methodology – Exploratory Data Analysis (Cont.)

Box Plots

Boxplot of x_7



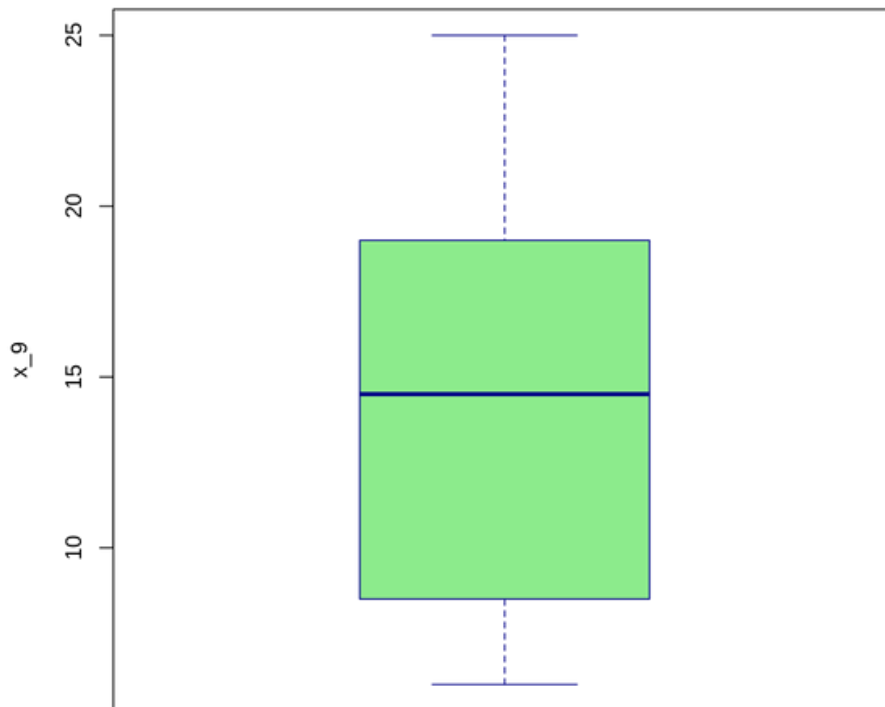
Boxplot of x_8



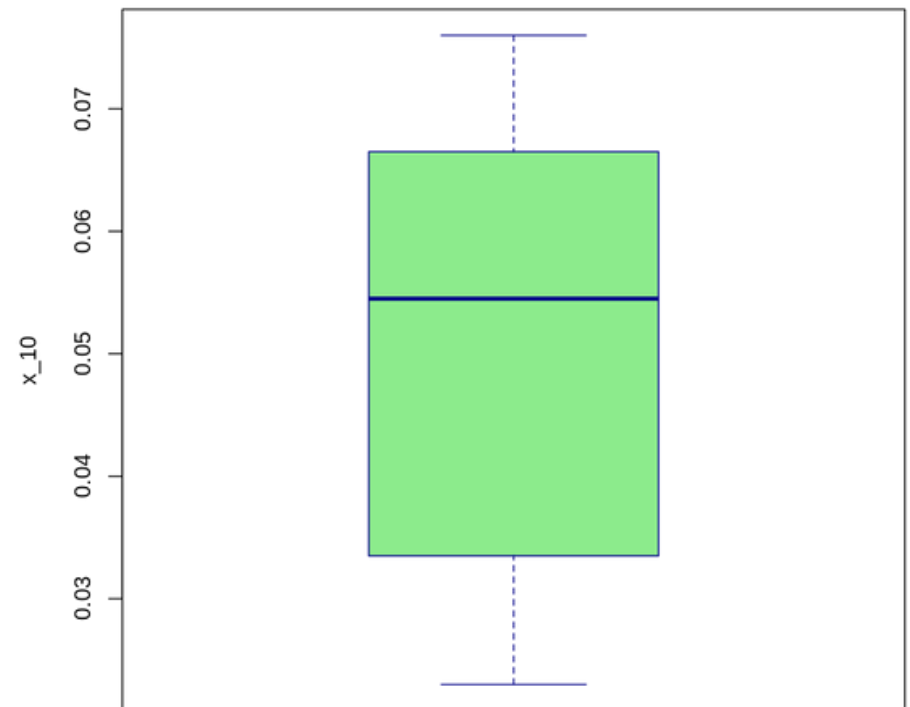
Methodology – Exploratory Data Analysis (Cont.)

Box Plots

Boxplot of x_9



Boxplot of x_10



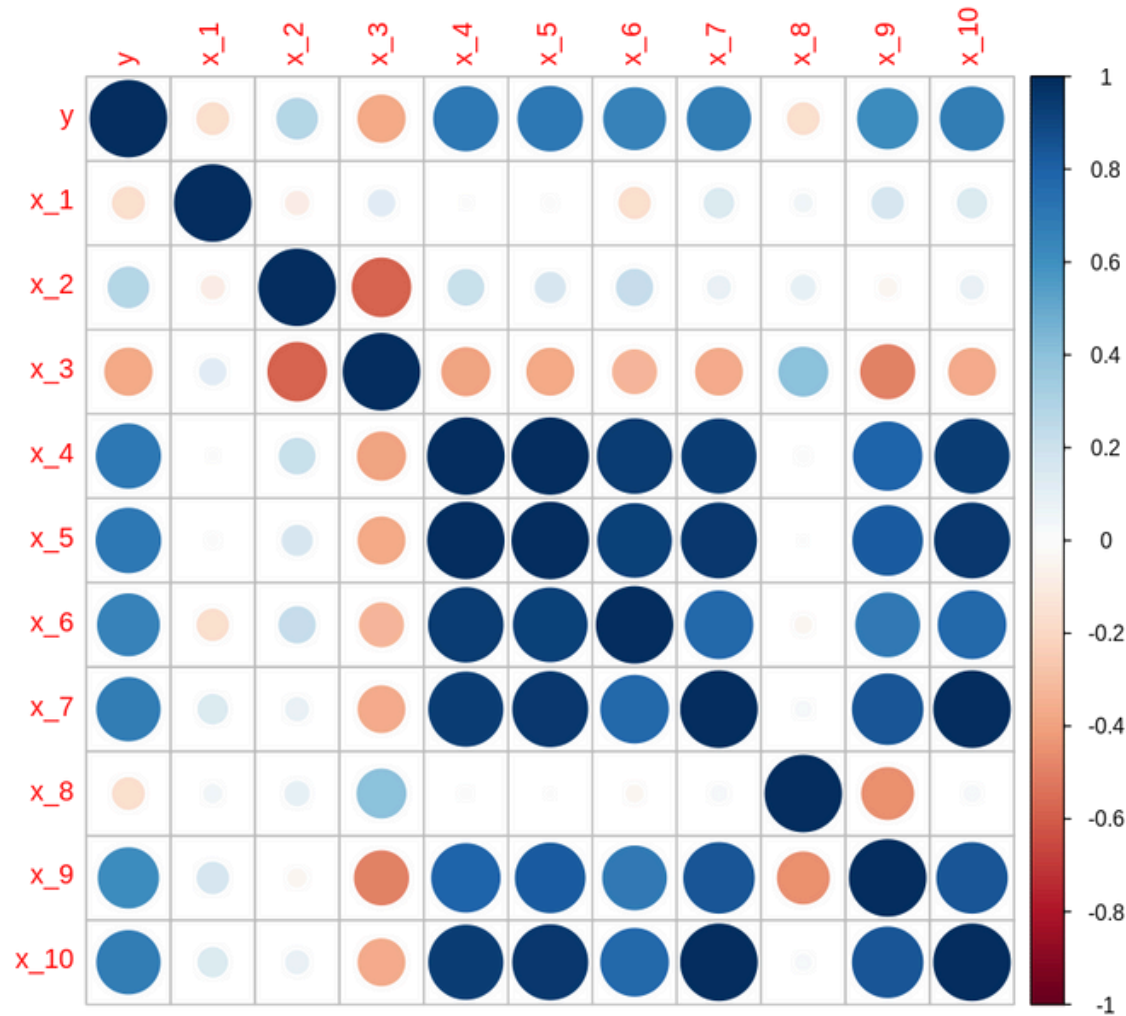
Methodology – Exploratory Data Analysis (Cont.)

Coefficient of Variation: To assess the relative variability of features irrespective of their units and scale

Variables	Coefficient of Variation (CV)
y	11.535699
x_1	89.602867
x_2	3.174758
x_3	54.825594
x_4	31.910174
x_5	34.058630
x_6	37.844245
x_7	34.829570
x_8	20.283178
x_9	39.850510
x_{10}	34.829570

Methodology – Exploratory Data Analysis (Cont.)

Correlation Plot: To identify linear relationships and detect multicollinearity among variables



Methodology – Regression Approach

Initial Approach: Ordinary Least Squares (OLS) Regression

Purpose:

- To build a simple, interpretable baseline for wine quality prediction

Key Assumptions:

- *Linearity* – Predictors relate linearly to the response
- *Normality of Errors* – Residuals follow a normal distribution
- *Homoscedasticity* – Constant variance of residuals
- *Independence* – Error terms are independent
- *No Multicollinearity* – Predictors not highly correlated (partially violated already)

Mathematical Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon$$

where,

- y : *response variable*
- $x_1, x_2, x_3, \dots, x_p$: *predictor variables*
- β_0 : *intercept*
- $\beta_1, \beta_2, \beta_3, \dots, \beta_p$: *regression coefficients*
- ε : *random error*

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression

Why Use OLS Despite Multicollinearity?

- Provides a clear, interpretable benchmark
- Helps evaluate performance gains from advanced models

Outcome:

Model 1.1 summarises regression coefficients, indicating the direction and strength of feature influence on wine quality

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression: Model 1.1 Summary

Residuals:

	Min	1Q	Median	3Q	Max
	-1.8138	-0.6850	0.1717	0.4833	1.9544

Coefficients: (2 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9.120845	17.561559	-0.519	0.610
x_1	-0.855781	0.702727	-1.218	0.240
x_2	6.310018	4.263064	1.480	0.157
x_3	0.014215	0.009461	1.502	0.151
x_4	-0.770508	2.593885	-0.297	0.770
x_5	3.386685	3.643216	0.930	0.366
x_6	-1.479912	2.169645	-0.682	0.504
x_7	NA	NA	NA	NA
x_8	-13.450969	8.700101	-1.546	0.140
x_9	-0.125294	0.246570	-0.508	0.618
x_10	NA	NA	NA	NA

Residual standard error: 1.208 on 17 degrees of freedom

Multiple R-squared: 0.7201, Adjusted R-squared: 0.5883

F-statistic: 5.466 on 8 and 17 DF, p-value: 0.001612

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression

Multicollinearity Issue: Model 1.1

► **Coefficients for x_7 and x_{10} marked as “not defined due to singularities”:**
Indicates perfect multicollinearity — some predictors are exact linear combinations of others

► **Action Taken:**

Performed Alias Structure Analysis on the training dataset

► **Purpose of Alias Check:**

To detect redundant predictors causing singularities in the design matrix

► **Result:**

Model :

$y \sim x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10}$

Complete :

	(Intercept)	x_1	x_2	x_3	x_4	x_5	x_6	x_8	x_9
x_7	0	0	0	0	0	1	-1	0	0
x_{10}	0	0	0	0	0	1/50	-1/50	0	0

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression

Multicollinearity Issue: Solution Implemented

- Removed redundant predictors (x_7 and x_{10}) from the training dataset
- Refit the model (**Model 1.2**) using Ordinary Least Squares (OLS) without multicollinear variables

Residuals:

Min	1Q	Median	3Q	Max
-1.8138	-0.6850	0.1717	0.4833	1.9544

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9.120845	17.561559	-0.519	0.610
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Residual standard error: 1.208 on 17 degrees of freedom

Multiple R-squared: 0.7201, Adjusted R-squared: 0.5883

F-statistic: 5.466 on 8 and 17 DF, p-value: 0.001612

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression

Model Assessment: Model 1.2

- Adjusted R-squared = 0.5883, indicating moderate explanatory power
- None of the individual predictors are statistically significant
- Large standard errors and weak predictor significance suggested persistent multicollinearity issues
- To investigate further, Variance Inflation Factor (VIF) scores were computed for Model 1.2

VIF scores for Model 1.2

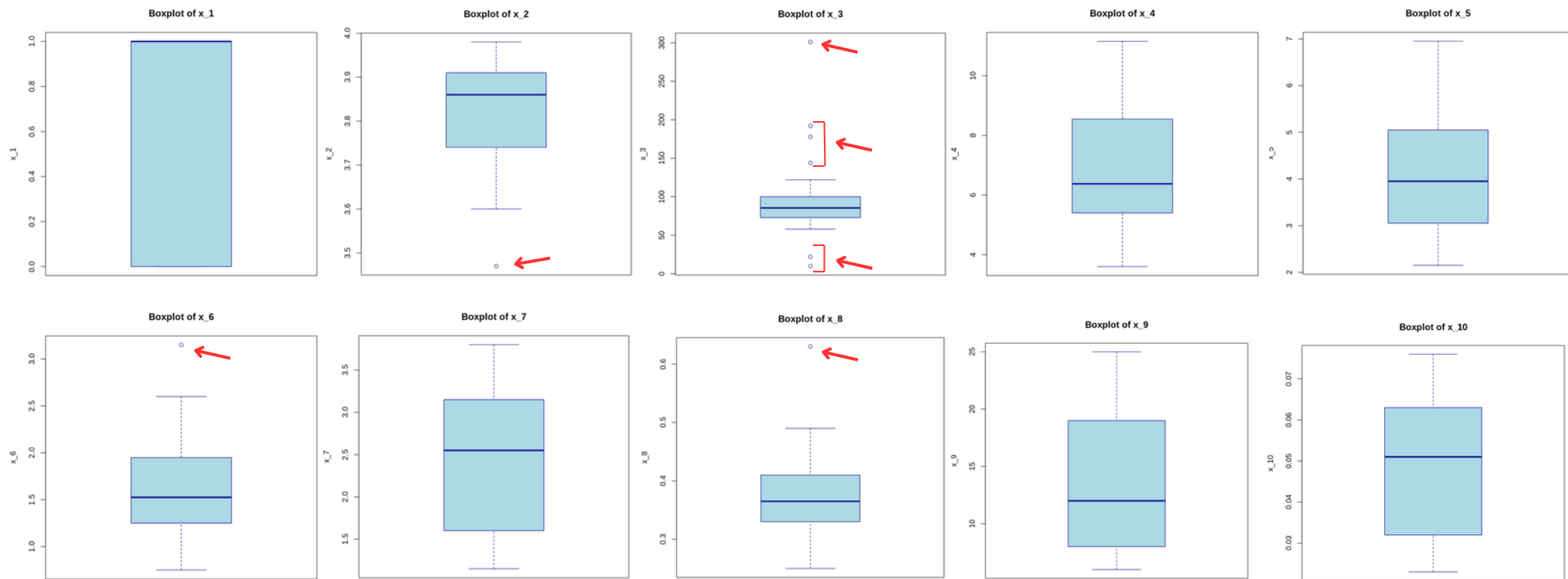
x_1	x_2	x_3	x_4	x_5	x_6	x_8
2.149156	5.339411	4.983590	520.757328	424.374295	25.588846	8.639865
x_9						
36.351670						

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression

★ Training data likely inherited outliers from the original dataset

↳ Box plots were drawn to visually identify the presence of outliers



Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression: Outlier Handling

Problem: Presence of Outliers can impact model performance and interpretability

Remedy: After resolving singularities, **Model 1.3** was fitted by excluding identified outliers (summary given below)

Aim: To improve model fit and coefficient stability

Summary:

Residuals:

Min	1Q	Median	3Q	Max
-1.6543	-0.4275	-0.0358	0.5970	1.7735

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.077758	18.863744	0.110	0.9139
x_1	-0.124129	0.835945	-0.148	0.8841
x_2	4.451546	4.442579	1.002	0.3333
x_3	0.009551	0.009751	0.979	0.3440
x_4	-1.657373	2.685038	-0.617	0.5470
x_5	5.629507	4.050116	1.390	0.1862
x_6	-1.925880	2.191682	-0.879	0.3944
x_8	-21.929684	10.349872	-2.119	0.0525 .
x_9	-0.361654	0.285718	-1.266	0.2263

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.194 on 14 degrees of freedom

Multiple R-squared: 0.7364, Adjusted R-squared: 0.5858

F-statistic: 4.89 on 8 and 14 DF, p-value: 0.004848

Methodology – Regression Approach (Cont.)

Ordinary Least Squares (OLS) Regression: Outlier Handling

Model Comparison:

- In Model 1.2, presence of high multicollinearity was noticed
- In Model 1.3, after dropping outliers no significant improvement was observed

Next Step:

- Address the issue of multicollinearity identified in the previous models
- Explore regression techniques that incorporate feature selection

Methodology – Regression Approach (Cont.)

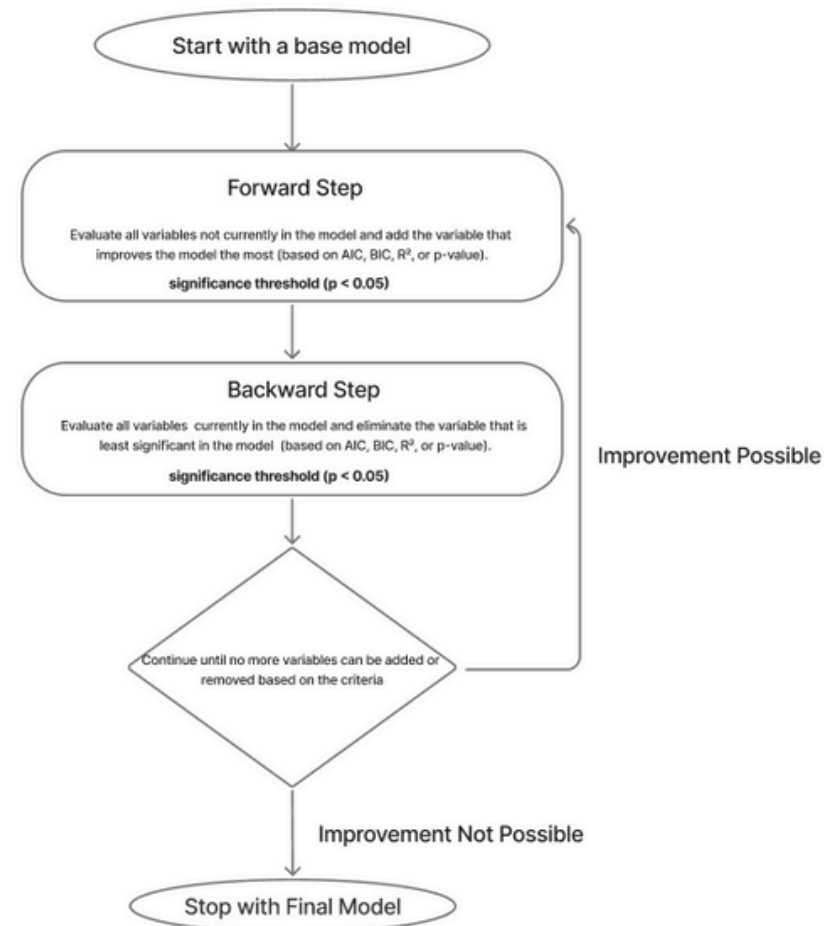
Feature Selection Approach: Stepwise Regression (Bidirection Model)

Why Use Stepwise Regression?

- Provides a feature selection technique benchmark
- Helps evaluate performance gains from advanced models

Why Not Using L1/L2 Regularization?

- **Presence of outliers:** Training dataset is affected by outliers
- **Sensitivity to Outliers:** Ridge and Lasso rely on least squares loss, making them highly sensitive to outliers
- **Risk of Information Loss:** Removing outliers to enable regularisation would have led to a 26.9% data loss



Methodology – Regression Approach (Cont.)

Stepwise Regression (Bidirection Model)

Outcome:

Model 2.1 summarises regression coefficients, indicating the direction and strength of feature influence on wine quality

Summary:

Residuals:

Min	1Q	Median	3Q	Max
-1.92704	-0.48727	-0.07742	0.56145	2.35927

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.724561	9.883837	-0.782	0.4436
x_1	-0.768065	0.470100	-1.634	0.1179
x_2	5.336196	2.591570	2.059	0.0527 .
x_3	0.013476	0.007299	1.846	0.0797 .
x_5	1.202803	0.202741	5.933	8.4e-06 ***
x_8	-8.593010	3.762480	-2.284	0.0334 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.163 on 20 degrees of freedom
Multiple R-squared: 0.6947, Adjusted R-squared: 0.6184
F-statistic: 9.101 on 5 and 20 DF, p-value: 0.0001227

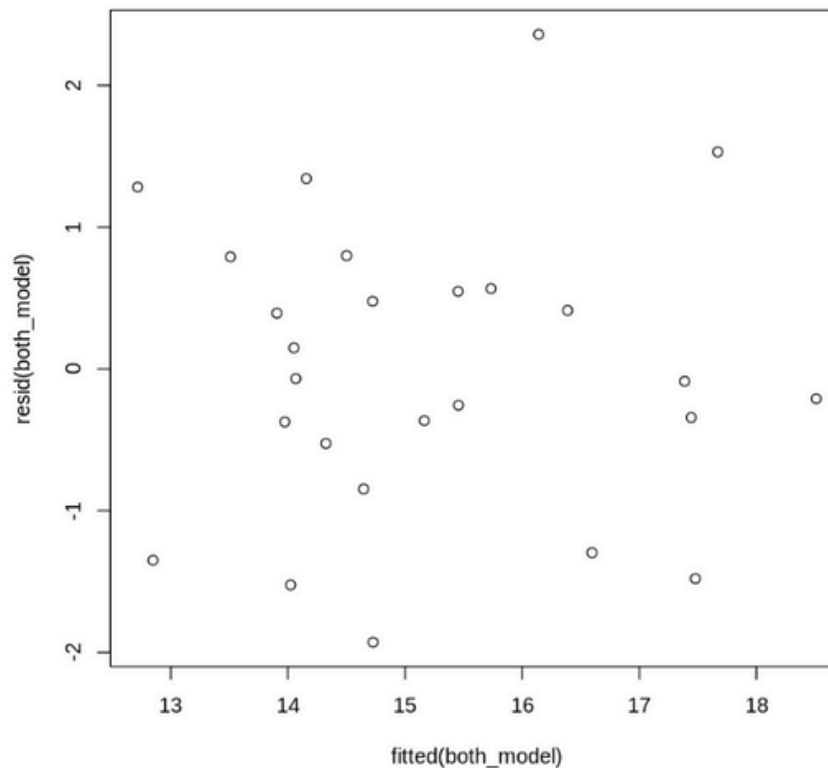
Methodology – Regression Approach (Cont.)

Stepwise Regression (Bidirection Model)

heteroscedasticity Issue: Model 2.1

- Residual standard error remained high (1.163) despite addressing multicollinearity suggests potential presence of heteroscedasticity
 - ↳ Residuals vs. fitted plot was analyzed to check for homoscedasticity

Plot:



Methodology – Regression Approach (Cont.)

Stepwise Regression (Bidirection Model)

Current Secnario:

- Multiple OLS assumptions violated: multicollinearity, heteroscedasticity, and outlier sensitivity

Next Step:

- Shifted focus to more robust modelling techniques that relax strict assumptions

Methodology – Regression Approach (Cont.)

Robust Regression (M Estimators):

Introduced Robust Regression using Huber's Loss Function as the foundation of our new modelling strategy

Huber's Loss Function:

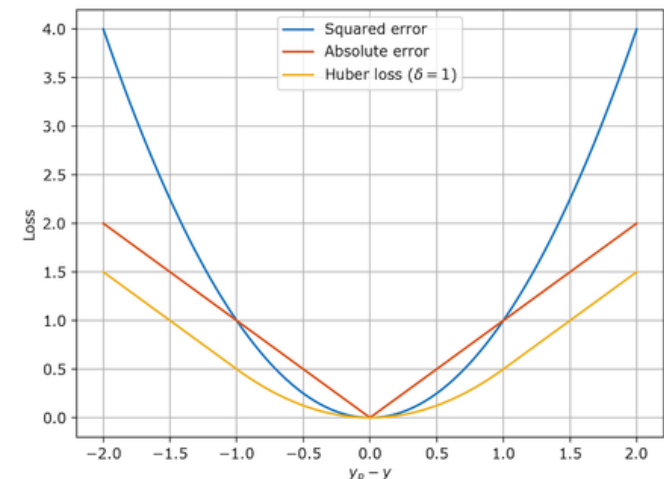
- Huber's Loss Function (1964) blends squared loss (OLS) and absolute loss (LAD) for robustness against outliers
- Acts quadratically for small residuals and linearly for large residuals.
- Forms a type of M-estimator, minimizing a transformed sum of residuals for robust parameter estimation

Mathematically, the Huber loss function is defined as:

$$L_{\delta}(r) = \begin{cases} \frac{1}{2}r^2 & \text{for } |r| \leq \delta \\ \delta(|r| - \frac{1}{2}\delta) & \text{for } |r| > \delta \end{cases}$$

where, $r = y_i - \hat{y}_i$: residual;

δ : hyperparameter [tuning parameter that determines the cutoff between quadratic and linear behaviour]



Methodology – Regression Approach (Cont.)

Robust Regression (M Estimators):

Transition to Robust Regression Model with Huber's Loss (Model 3.1)

- Model 3.1 is obtained by replacing the MSE loss of Model 1.2 with Huber's loss function
- Aims to achieve improved robustness against outliers and model instability

Summary:

Residuals:

Min	1Q	Median	3Q	Max
-2.0143	-0.6068	0.1272	0.4721	2.2530

Coefficients:

	Value	Std. Error	t value
(Intercept)	-8.2084	19.4618	-0.4218
x_1	-0.9339	0.7788	-1.1992
x_2	6.1926	4.7243	1.3108
x_3	0.0149	0.0105	1.4258
x_4	-0.4763	2.8746	-0.1657
x_5	2.8197	4.0374	0.6984
x_6	-1.4506	2.4044	-0.6033
x_8	-14.5551	9.6415	-1.5096
x_9	-0.1134	0.2732	-0.4151

Residual standard error: 0.7845 on 17 degrees of freedom

Methodology – Regression Approach (Cont.)

Robust Regression (M Estimators):

Limitations of Model 3.1 (Linear Regression with Huber's):

- Predictors are not statistically significant at standard significance levels
- R^2 (0.7133) and Adjusted R^2 (0.5783) indicate moderate model fit with signs of overfitting or collinearity
- Despite lower residual error (0.7845), multicollinearity persists

What's Next?

- New M estimator adapted with high resistance to outliers with stability under multicollinearity and heteroscedasticity

Methodology – Regression Approach (Cont.)

Robust Regression (MM Estimator):

MM Estimator

- Developed by Prof. Yohai, the MM Method combines S-estimators (for robustness) and M-estimators (for efficiency)
- Designed to handle outliers, heteroscedasticity and multicollinearity (moderately) simultaneously
- Has a high breakdown point (up to 50%), meaning it remains reliable even with a large proportion of outliers
- Applies multi-stage reweighing (robust scale → robust coefficients → refined coefficients) for better stability

Methodology – Regression Approach (Cont.)

Robust Regression (MM Estimator):

Why MM Estimator over Huber's Loss?

- Huber's method moderately resists outliers but struggles with high-leverage outliers and collinearity
- MM Estimator handles non-constant variance more flexibly and avoids instability from collinearity
- Provides more accurate and stable estimates compared to the single reweighting of Huber's Loss

Outcome:

```
Residuals:
    Min       1Q   Median       3Q      Max
-4.8231 -0.5730  0.1021  0.3658  2.2518

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.609163    9.845506  -0.773   0.4502
x_1          -0.581110    0.819926  -0.709   0.4881
x_2           4.617232    2.616750   1.764   0.0956 .
x_3           0.008216    0.009920   0.828   0.4190
x_4          -2.086893    1.959675  -1.065   0.3018
x_5           4.093641    2.234675   1.832   0.0845 .
x_6          -0.190132    1.629617  -0.117   0.9085
x_8           4.706333   12.711890   0.370   0.7158
x_9           0.080280    0.166883   0.481   0.6366
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Robust residual standard error: 1.003
Multiple R-squared:  0.7278,    Adjusted R-squared:  0.5997
Convergence in 18 IRWLS iterations
```

Methodology – Regression Approach (Cont.)

Robust Regression (MM Estimator):

Model Assessment: Model 1.2

➤ Good model fit:

- Multiple $R^2 = 0.7278$, Adjusted $R^2 = 0.5997 \rightarrow$ about 60% of variability explained
- Robust residual standard error = 1.003 $\rightarrow$ indicates good pattern capture despite outliers

➤ Predictor insights:

- x_2 and x_5 show marginal significance at the 10% level with positive influence
- Other predictors ($x_1, x_3, x_4, x_6, x_8, x_9$) are not statistically significant, suggesting weak or uncertain effects

➤ Conclusion:

- Given the satisfactory variability explained, alongside knowledge barriers and time constraints, the project was not escalated further at this stage

Model Fitting

- After fitting different multiple regression models, we compared them to find the best-fitted model based on Adjusted R^2
- The models differ by estimation techniques and robustness to outliers and multicollinearity

Model No.	Model Description	Adjusted R^2
Model 1.2	OLS Regression	0.5883
Model 1.3	OLS Regression (after dropping outliers)	0.5858
Model 2.1	Bothways Stepwise Linear Regression	0.6184
Model 3.1	Robust Regression (Huber's Loss)	0.5783
Model 3.2	Robust Regression (MM Estimation via <code>lmrob</code>)	0.5997

Model Comparison

Best-Fitted Model Selection

- Bothways Stepwise Linear Regression had the highest Adjusted R^2 (0.6184), but violated homoscedasticity assumption, affecting estimate's reliability
- Robust Regression (MM-estimation via Imrob) achieved an Adjusted R^2 of 0.5997, offering stable and robust estimates even with outliers

Selected Model:

- Robust Regression (MM-estimation via Imrob) — best balance between fit and robustness
- **Mathematical Model:**

$$y = -7.6092 - 0.5811x_1 + 4.6172x_2 + 0.0082x_3 - 2.0868x_4 + 4.0936x_5 - 0.1901x_6 + 4.7036x_8 + 0.082x_9$$

Cross Validation

► Objective:

- Assess generalisability and predictive accuracy using a separate test dataset (unseen data)

► Method:

- Applied previously fitted models to the test set and compared their predictive performance

► Metric Used:

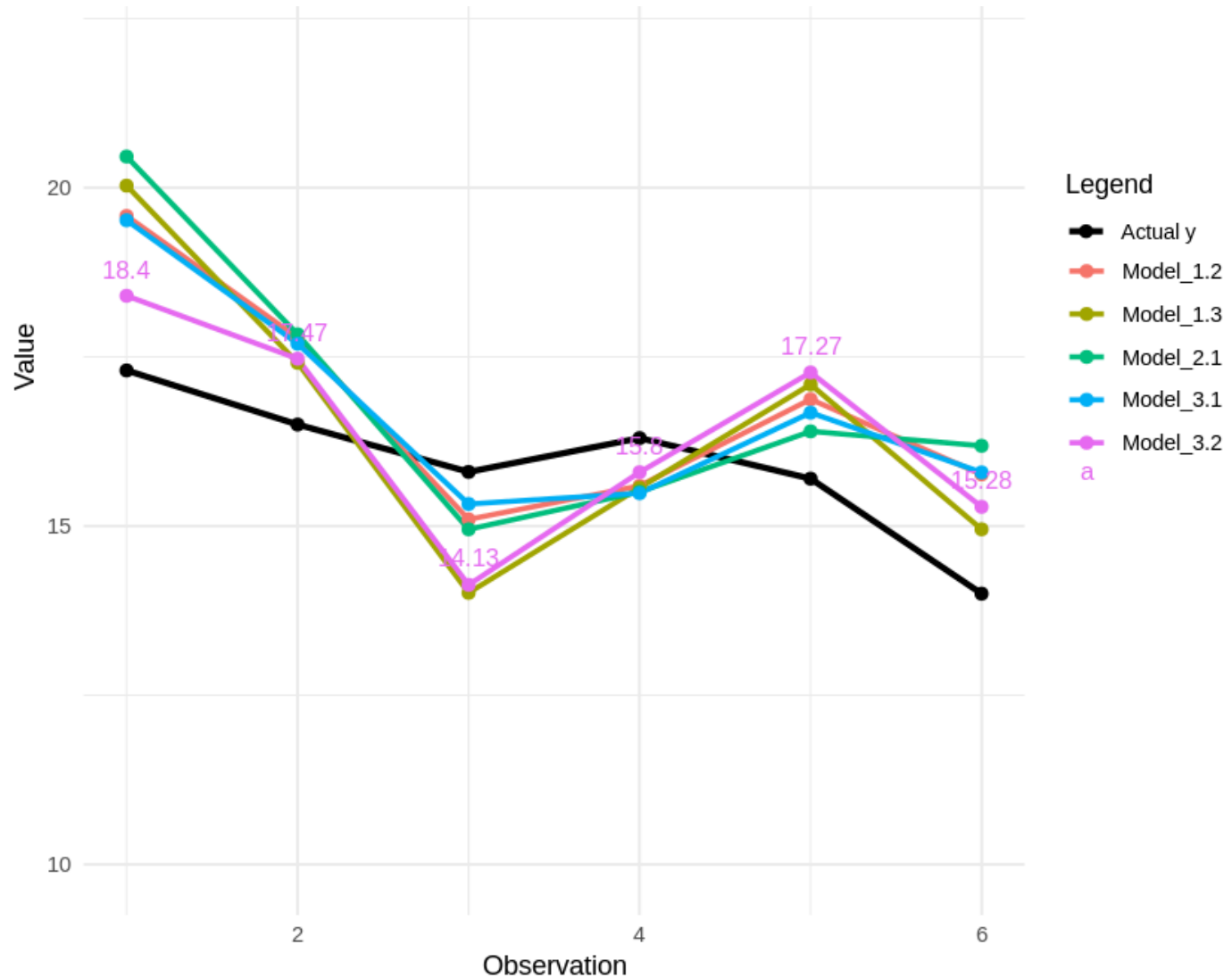
- Root Mean Squared Error (RMSE) on test data to evaluate fit and predictive power

► Purpose:

- Identify the model that best balances goodness of fit, robustness, and generalizability

Cross Validation (Cont.)

► Comparative Results: (Actual vs Predicted Plot)



Cross Validation (Cont.)

- The predictive performance of each model was evaluated using Root Mean Squared Error (RMSE) on test data, as summarised below

Model No.	Model Type	RMSE
Model: 1.2	OLS Model	1.434063
Model: 1.3	OLS Model after dropping outliers	1.574437
Model: 2.1	Bothway Stepwise Regression Model	1.750833
Model: 3.1	Robust Regression with M estimator (Huber's loss function)	1.378922
Model: 3.2	Robust Regression with MM estimator	1.246141 ★

Conclusion

➤ Explored wine quality prediction using a small, multicollinear, and outlier-affected dataset

- OLS Regression revealed instability due to multicollinearity and outliers
- Stepwise Regression improved model simplicity but suffered from heteroscedasticity
- Adopted Robust Regression Techniques:
 - Huber's M-estimator → Mitigates the outliers' effect. but failed to resolve heteroscedasticity and multicollinearity
 - MM-estimator → Most effective, with lowest RMSE and high Adjusted R^2

✅ **MM Method is the best predictive model obtained within the limitations of our study.**

The project highlights the importance of robust, assumption-resilient models in real-world, small-sample data analysis—especially in fields like oenology, where objectivity in quality assessment is vital

Limitations

Small Dataset (n = 32):

- Limits the use of data-hungry methods like neural networks or ensemble models; reduces the statistical power of classical regression

Multicollinearity Remains:

- MM Method handled outliers well but did not inherently resolve multicollinearity

Alternative Techniques Not Used:

- Dimensionality reduction methods like PCA and PLS could help improve model interpretability and reduce multicollinearity

Limited Validation Strategy:

- Used only an 80:20 train-test split; k-fold cross-validation could improve generalizability and performance assessment

Unexplored Robust Ridge Regression:

- Initially considered but not pursued due to time and familiarity constraints; remains a promising future direction

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- **Data Source:**

- UCI Machine Learning Repository

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- **Books:**

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- Robust Statistics: The Approach Based on Influence Functions. [Wiley] - Hampel, F. R., et al. (1986)
- Generalized Linear Models: With Applications in Engineering and the Sciences [Wiley] - Myers, R.H.; Montgomery, D.C.; Vining, G.G.; Robinson, T.J

Thank You!